Meta-Intelligence — A Cross-Domain Comparative Framework under Definition

Abstract

Inspired by the lever, this paper proposes a reference system for intelligence that enables dimensionalized expression for cross-domain comparison. By defining baseline structure and baseline energy, an intelligence zero-point scale is established, with *Archi* as the provisional unit of intelligence. The proposed comparative framework remains at the definitional level and has not yet achieved complete closure of cross-domain physical quantities. This methodology is an exploratory physical manipulation intended to provide a reference framework and open a preliminary path for future cross-domain quantification of intelligence.

1. Introduction

Human civilization began with the use of sticks; the lever may be the first "intelligent structure" in history—an artificial system that alters energy behavior through structure. Based on this heuristic "zero-point structure," this paper attempts to construct a physical dimensional reference system for intelligence. By defining baseline structure and baseline energy, an intelligence zero-point scale is established, enabling cross-domain observation and comparison of different structural systems under a unified baseline.

2. Baseline task and Baseline structure

2.1 Baseline Task

The baseline task is defined as moving a 1 kg mass horizontally by 1 m. The energy consumption is determined solely by sliding friction, independent of gravitational potential or environmental factors, thus serving as a unified baseline across structures.

2.2 Baseline Structure: Why the Lever

The baseline Structure is defined as a 1:1 lever with sliding contact. The lever is chosen as the zero-point structure for intelligence scale because:

- *Minimal structural intervention*: A 1:1 lever alters energy pathways without amplifying force, distance, storing, or converting energy.
- *Computable energy behavior*: Energy consumption depends only on sliding friction and geometry, ensuring repeatability and measurability.
- *Cross-domain comparability*: Lever energy behavior can be compared with wheels, engines, microcontrollers under the same task.
- *Structural independence: The function of the lever is determined by the structure itself, rather than by the user's strength or skill, in accordance with the definition that "intelligence is the unfolding of energy under structural constraints." That is, the intelligence equation:*

$$I = dE/dt|\text{结构约束}$$

where I is intelligence, E is energy. This equation was first proposed in *Toward a Physics of Intelligence: A Unified Framework of Energy, Mass, and Intelligence* to describe intelligence as the unfolding of energy under structural constraints.

Thus, the lever is established as the zero-point structure of intelligence. This choice is not arbitrary but deliberately grounded in natural law.

2.3 Baseline Energy: Source of Intelligence Dimension

Using Coulomb's law of kinetic friction:

$$\boldsymbol{F_{\text{friction}} = \mu_k N = \mu_k mg}$$

where:

- $F_{friction}$: sliding friction force (N)
- μ_k: coefficient of kinetic friction (dimensionless)
- m: mass (kg)
- $g=9.81\,m/s^2$: gravitational acceleration

The energy required for the baseline task:

$$\boldsymbol{E_0 = F_{\text{friction}} L = \mu_k mgL}$$

Where:

- E_0 is the energy required for the baseline task (unit: J),
- m = 1 kg,
- L = 1 m,
- g = 9.81 m/s²,

- μ_k is the coefficient of kinetic friction, with typical values 0.3 – 0.6 (dimensionless).

The significance of this energy E_0 is as follows:

2.3.1 Dimensional closure

Intelligence defined as:

$$\boldsymbol{I = \frac{1}{E_0}}$$

with dimension J^{-1}.

In order for intelligence to become a measurable physical quantity, a baseline energy E_0 must be introduced as the zero point of the scale.

2.3.2 Minimality

The baseline structure of the lever intervenes in energy behavior in the most minimal way: it only changes the direction of force, without altering the energy pathway or introducing additional structural complexity. Under this minimal structural constraint, the energy required for the lever to accomplish the baseline task is denoted as E_0, and it is selected as the baseline energy for the intelligence scale.

2.3.3 Reference frame

All structural systems can be comparatively observed under the same baseline Energy E0, thereby forming a cross-domain reference framework for intelligence. This framework does not provide a unified quantitative result, but rather offers a comparable observational basis for the performance of different structural systems. On this basis, we provisionally define the unit of intelligence as Archi:

$$1\ Archi = \frac{1}{E_0}$$

Where:

- E0: the baseline energy of the baseline task.
- The dimension of intelligence is therefore J^{-1}, the reciprocal of energy.

Thus, the intelligence scale is essentially an energy ratio, with its unit defined as Archi.

3. Wheel: First Leap in Structural Intelligence

The wheel shifts the energy pathway from sliding friction to rolling resistance, and its energy consumption is:

$$E_{\text{wheel}} = c_{rr}\, mgL$$

where crr is the coefficient of rolling resistance, with typical values ranging from 0.005 to 0.02 (dimensionless).

The relative intelligence between the wheel and the lever is:

$$\frac{I_{\text{wheel}}}{I_{\text{lever}}} = \frac{\mu_k}{c_{rr}}$$

Taking typical ranges:

- Coefficient of sliding friction μ_k: 0.3 – 0.6
- Coefficient of rolling resistance c_{rr}: 0.005 – 0.02

All combinations yield the following ratios:

$$\{\frac{0.3}{0.02} = 15,\ \frac{0.6}{0.02} = 30,\ \frac{0.3}{0.005} = 60,\ \frac{0.6}{0.005} = 120\}$$

Therefore, the range of intelligence improvement of the wheel relative to the lever is:

$$\frac{I_{\text{wheel}}}{I_{\text{lever}}} \in [15,\ 120]$$

It is evident that the wheel achieves an intelligence leap of 1–2 orders of magnitude compared to the lever, with an improvement of 15–120 times.

4. Intelligence of Power Structures

For engines, intelligence can be defined as:

$$I_{\text{engine}} = \frac{P}{E_0}$$

where P is the output power of the engine (unit: W = J/s). This expression represents the number of "baseline task equivalents" the engine can accomplish per second.

Example: Quantifying the intelligence of a 15-horsepower diesel engine

15 horsepower (hp) converted into SI power units:

$$1\,hp = 735.5\,W$$

Therefore:

$$P = 15 \times 735.5 = 11032.5\,W$$

The baseline energy of the baseline task is:

$$E_0 = \mu_k mgL$$

Taking typical values:

- μ_k=0.5
- m=1 kg
- L=1 m
- g=9.81 m/s2

Then:

$$E_0 = \mu_k\, mgL = 0.5 \times 1 \times 9.81 \times 1 = 4.905\,\text{J}$$

Thus, the intelligence of the 15-horsepower diesel engine is:

$$I_{engine} = \frac{11032.5}{4.905} \approx 2248\ Archis/s$$

This provides a measurement of the intelligence scale corresponding to the power output capability of a 15-horsepower diesel engine.

5. Intelligence of Information Structures

The fundamental capability of an information structure can be expressed as the number of operations completed per unit of energy (operations per joule):

$$\frac{operations}{J} = \frac{operations/s}{P}$$

where P is the system’s power consumption (unit: W = J/s).

Mapping this to the baseline scale gives the intelligence of an information structure:

$$I_{info} = \left(\frac{operations}{J}\right) E_0$$

This expression indicates the number of operations an information structure can perform when consuming energy equivalent to the baseline task E0, thereby allowing information processing capability to be compared with mechanical structural capability on the same intelligence scale.

Example: Intelligence of the 8051 MCU

Taking the classic 8051 microcontroller as an example:

- Clock frequency: 12 MHz
- Instruction throughput: 1 M operations/s (≈ 12 cycles per instruction)
- Supply voltage: 5 V
- Average power consumption: P≈20 mW=0.02 W

Operations per joule:

$$\frac{operations}{J} = \frac{operations/s}{P} = \frac{1 \times 10^6}{0.02} = 5 \times 10^7\ operations/J$$

Mapping to the baseline scale: baseline energy of the baseline task (previously defined):

$$E_0 = \mu_k mgL = 0.5 \times 1 \times 9.81 \times 1 = 4.905\,J$$

Thus, the intelligence of the 8051 information structure is:

$$I_{info_8051} = \left(\frac{operations}{J}\right) E_0 = 5 \times 10^7 \times 4.905 \approx 2.45 \times 10^8$$

This means that when consuming energy equivalent to completing one baseline task E0, the 8051 MCU can execute approximately 2.45×10^8 instructions, providing a measure of its computational intelligence.

For high-performance microprocessors, nE0 can be adopted as the reference energy for measurement and observation, thereby reflecting the intelligence scale of more complex tasks or larger-scale computations.

6. Cross-Domain Reference Observation

Since all structures take the baseline Energy E0 as the baseline, the intelligence scale can extend across different types of systems:

- Lever (zero-point structure)
- Wheel (structural optimization)
- Engine (power structure)
- 8051 MCU (information structure)

Thus, mechanical, energetic, and informational systems can be observed in a cross-domain reference framework. This framework does not provide a unified quantitative result, but rather establishes a comparable observational basis for the intelligence performance of different structural systems.

7. Discussion: From Scale Definition to the Dynamic Form of the Intelligence Equation

In Section 2.3.1, this paper presented the scale definition of intelligence:

$$I = \frac{1}{E_0}$$

This definition establishes the physical dimension of intelligence, allowing the organization of energy in different structures under the same task conditions to be observed within a unified reference frame. This form depicts the performance of structures along specific energy pathways and serves as the scaled expression of intelligence. It provides a cross-domain reference framework, enabling mechanical, energetic, and informational structures to be compared within the same system.

In a more general structure–energy framework, intelligence can be expressed as the unfolding of energy under structural constraints. This perspective is given by the intelligence equation:

$$\boldsymbol{I} = \frac{\boldsymbol{dE}}{\boldsymbol{dt}}\Big|_{\textit{结构约束}}$$

The intelligence equation characterizes the unfolding of energy under structural constraints, representing the dynamic structural form of intelligence.

The scale definition:

$$\boldsymbol{I} = \frac{\boldsymbol{1}}{\boldsymbol{E_0}}$$

can be regarded as the integral expression of the intelligence equation under specific tasks and specific energy pathways, used to establish dimensionality and reference baselines. Meanwhile, the intelligence equation provides a general structural dynamic description, revealing the unfolding of energy under structural constraints.

Therefore, the quantitative system of intelligence proposed in this paper can be understood as a concrete elaboration within the framework of the intelligence equation: the scale definition provides dimensionality and comparability, while the dynamic structural form provides the mechanism of evolution. Together, they constitute a unified physical description of intelligence.

8. Conclusion

This paper proposes a physical quantity system of intelligence based on the definition of the baseline task, enabling structures from different domains to be observed under a common scale. The scale takes the baseline task and baseline structure as the zero point,

with the energy pathway as its core, thereby making intelligence a physical quantity with unified reference. By establishing the dimensionality of intelligence through the lever, and further extending it to the wheel, the engine, and information-processing structures, mechanical, energetic, and informational systems can all be observed within the same physical framework through cross-domain reference.

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This document is a preprint draft. Academic exchange and discussion are welcome.